

Exercise 1:

Dependencies in pom.xml:

```
21 <dependencies>
22   <dependency>
23     <groupId>org.slf4j</groupId>
24     <artifactId>slf4j-api</artifactId>
25     <version>1.7.30</version>
26   </dependency>
27   <dependency>
28     <groupId>ch.qos.logback</groupId>
29     <artifactId>logback-classic</artifactId>
30     <version>1.2.3</version>
31   </dependency>
32 </dependencies>
```

ExampleLogger Code:

```
1  package com.example;
2
3  import org.slf4j.Logger;
4  import org.slf4j.LoggerFactory;
5
6  public class LoggingExample {
7      private static final Logger logger = LoggerFactory.getLogger(LoggingExample.class);
8
9      Run | Debug
10     public static void main(String[] args) {
11         logger.error("This is an error message.");
12         logger.warn("This is a warning message.");
13     }
14 }
```

Output:

```
PS C:\Users\anjan\OneDrive\Documents\College\Educational\Assignments\CSE\java\sl4j\logging-demo> mvn exec:java "-Dexec.mainClass=com.example.LoggingExample"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:logging-demo >-----
[INFO] Building logging-demo 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec:3.5.0:java (default-cli) @ logging-demo ---
00:39:02.902 [com.example.LoggingExample.main()] ERROR com.example.LoggingExample - This is an error message.
00:39:02.904 [com.example.LoggingExample.main()] WARN com.example.LoggingExample - This is a warning message.
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 0.599 s
[INFO] Finished at: 2026-07-03T00:39:02+05:30
[INFO] -----
```